MIL-G-60662B (AR)
01 May 1991
SUPERSEDING
MIL-G-60662A (AR)
30 June 1982

MILITARY SPECIFICATION

GENERATOR, SIGHT CURRENT 12000239

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements, quality assurance provisions and packaging for the Generator Sight Current 12000239, hereinafter referred to as the generator. The generator is part of Vulcan Weapon Systems M163A1 and M167Al (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-F-13926 Fire Control' Materiel, Manufacture and Inspection, General Specification for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1285

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MIL-I-45607	-	Inspection Equipment, AcquisitiOn, Maintenance and Disposition of
MIL-C-50785	-	Circuit Card Assembly, Sight Current Generator, General Specification for
MIL-C-50785/l	-	Circuit Card Assembly: 9337291 Voltage Comparator
MIL-C-50785/2	-	Circuit Card Assembly: 10545483 Firing Indicator
MIL-C-50785/7	-	Circuit Card Assembly: 10545704 Acquisition Time Delay
MIL-C-50785/8	-	Circuit Card Assembly: 10545705 Sight Correction
MIL-C-50785/9	-	Circuit Card Assembly: 10545706 Pulse Width Multiplier
MIL-C-50785/11	-	Circuit Card Assembly: 12000011 Mode Scaling
MIL-C-50785/12	-	Circuit Card Assembly: 12000012 Computation Control
MIL-C-50785/13	-	Circuit Card Assembly: 12000013 Atmospheric Correction
MIL-c-50785/14	-	Circuit Card Assembly: 12000010 Ballistic Constant
MIL-C-50785/15	-	Circuit Card Assembly: 12000015 Rate Series
MIL-C-50785/16	-	Circuit Card Assembly: 12000016 Range Series
MIL-C-50785/17	-	Circuit Card Assembly: 12000252 Time Delay

STANDARDS

MILITARY

MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-810		Environmental Test Methods and Engineering Guidelines
MIL-STD-2000	-	Standard Requirements for Soldering Electrical and Electronic Assemblies

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

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2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS (see 6.4)

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

9337291	Component Assembly, Voltage Comparator
10545126	Electromagnet, Range
10545170	Motor and Connector
10545260	Module, Torque Current Supply
10545270	Module, Magnet Current Supply
10545437	Potentiometer, Elevation Lead Angle
10545465	Isolator
10545483	Component Assembly, Firing Indicator
10545536	Inverter
10545704	Component Assembly, Acquisition Time Delay
10545705	Component Assembly, Sight Correction
10545706	Component Assembly, Pulse Width Multiplier
12000010	Component Assembly, Ballistic Constant
12000011	Component Assembly, Mode Scaling
12000012	Component Assembly, Computation Control
12000013	Component Assembly, Atmospheric Correction
12000015	Component Assembly, Rate Series
12000016	Component Assembly, Range Series
12000017	Harness Assembly
12000239	Sight Current Generator
12000242	Wiring Installation, Sight Current Generator
12000252	Component Assembly, Time Delay
12007500	20MM Self Propelled Air Defense Artillery Gun M163A1
12007501	20MM Towed, Air Defense Artillery Gun M167A1

PACKAGING DATA SHEET

PS12000239	Packaging of Sight Current Generator

(Copies of drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

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2.2 order of precedence. In the event of conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on Drawing 12000239, all associated drawings, and with all requirements specified in applicable specifications and standards. where applicable, the generator shall meet the requirements of MIL-F-13926 as follows:

Dimensions and tolerances
b. Inorganic protective surface finishings
c. Part identification and marking
d. Workmanship

The subassemblies of the generator shall have been inspected and accepted in accordance with their respective drawings and specifications prior to their installation in the generator.

The following subassemblies are covered by the indicated specifications:

Unit No.	Name	Specification
Al	Ballistic Constant 12000010	MIL-C-50785/14
A2	Mode Scaling 12000011	MIL-C-50785/11
A3	Computation Control 12000012	MIL-C-50785/12
A4	Voltage Comparator 9337291	MIL-C-50785/l
A5	Firing Indicator 10545483	MIL-C-50785/2
A6	Voltage Comparator 9337291	MIL-C-50785/l
A7	Acquisition Time Delay 10545704	MIL-C-50785/7

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Unit No.	Name	Specification
A8	Sight Correction 10545705	MIL-C-50785/8
A9	Pulse Width Multiplier 10545706	MIL-C-50785/9
A10	Voltage Comparator 9337291	MIL-C-50785/l
All	Time Delay 12000252	MIL-C-50785/17
A12	Atmospheric Correction 12000013	MIL-C-50785/13
Al 3	Magnet Current Supply 10545260	SQAP 10545270
A14	Torque Current Supply 10545260	SQAP 10545260
A16A1	Rate Series 12000015	MIL-C-50785/15
A16A2	Range Series 12000016	MIL-C-50785/16

3.3 Environmental service conditions.

3.3.1 Vibration. The generator shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to sinusoidal vibrations, in each of three mutually perpendicular axes, as follows:

Frequency Range	Amplitude
10 to 15 hertz	0.06 inches double amplitude
15 to 25 hertz	0.04 inches double amplitude
25 to 55 hertz	0.02 inches double amplitude

The frequency of vibration shall be swept logarithmically.
The cycling time shall be 120 minutes per axis.
The sweep rate shall be 3.75 minutes ± 10% per decade.
The 5-55-5 hertz sweep time is 3 minutes 45 seconds ± 10%.

3.3.2 Shock. The generator shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to a total of 18 impulse shocks consisting of three shock impulses of 15 g's ± 10%, 11 milliseconds ± 10%, half-sine wave applied in each direction of three mutually perpendicular axes.

3.3.3 Altitude.

3.3.3.1 Operating. The generator shall meet the requirements of 3.4 and 3.5 during and after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 inches of Hg).

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3.3.3.2 Non-operating. The generator shall meet the requirements of 3.4 and 3.5 after exposure for not less than 10 minutes to barometric pressure encountered at 40,000 feet above sea level (5.5 inches of Hg).

3.3.4 Temperature.

3.3.4.1 Operating. The generator shall meet the requirements of 3.4 and 3.5 when stabilized thermally at any temperature between -40 degrees F and +155 degrees F.

3.3.4.2 Non-operating (storage). The generator shall not suffer permanent damage when subjected to the constant temperature extremes of -60 degrees F for a period of 24 hours and +160 degrees F for a period of 4 hours (after temperature stabilization), and shall meet the requirements of 3.4 and 3.5 when returned to room ambient.

3.3.5 Humidity (non-operating) The generator shall meet the requirements of 3.4 and 3.5 after exposure to an atmosphere with a relative humidity in excess of 95 percent for a period of 10 days.

3.3.6 Rain. The generator shall be capable of operation after being subjected to Procedure I of MIL-F-13926A(MU), dated 5 November 1962.

3.4 Electrical. Each conductor shall be electrically continuous as specified by Runs Lists RL12000242 and RL12000017 of the generator. Continuity shall be indicated by a measurement of 0.5 ohms maximum for wires.

3.5 Performance. when the generator is provided with electrical power and with data for air density, muzzle velocity, angular position, and target Range and Range Rate, it shall compute required magnet and torque current for output to a lead-computing sight as specified in the following subparagraphs.

3.5.1 Functional characteristics.

3.5.1.1 Primary performance characteristics.

3.5.1.1.1 Magnet current generation. The generator shall be capable of providing a controlled direct-current (de) magnet current of variable magnitude, 100 milliamperes to 700 milliamperes, to the sight magnet coil.

3.5.1.1.2 Magnet current equation. The generator shall supply the magnet current required to implement the specified sight sensitivity by computing the required ^{V}IM which is the voltage present at the input of the magnet current supply module. The following equations for calculating ^{V}IM shall be solved by the generator. The equation terms are defined in Table I.

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3.5.1.1.2.1 Magnet current - target inbound - radar mode

$$V_{IM} = A_M \left[\left(\frac{-.0236L - 13.188}{R} + \frac{149314}{R^2} - \frac{8.8315}{R^3} \right) + R \left(\frac{.168}{R} - \frac{.4308}{R^2} \right) \right] - 3.938A_D + .15$$

NOTE : L = lamda in formulas of 3.5.1.1.2.1, 3.5.1.1.2.2, 3.5.101.4.1, 3.5.1.2.2.3.3.2, 3.5.1.2.3.9, 3.5.1.2.3.10, 3.5.1.2.4.3c(3).

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TABLE 1. Ballistic equation terms

Term	Definition
L	lamba
V_{IM}	Voltage at Input to Magnet Current Supply Module
L^2	$(L^2_E + L^2E) \times .866$
L^2_E	(Elevation Lead Angle)$^2 = \frac{O_A^2}{25.43}$ where O_A is the elevation angle between the gun line and the antenna
L^2_T	(Traverse Lead Angle)$^2 = \frac{O_B^2}{25.43}$ where O_B is the traverse angle between the gun line and the antenna
A_M	Muzzle Velocity:

Switch position	Muzzle velocity (ft/see)	AM (Gain factor)
1	3580	-1.905
2	3530	-1.889
3	3480	-1.873
4	3430	-1.857
5	3380	-1.841
6	3330	-1.825
7	3280	-1.809
8	3230	-1.792
9	3180	-1.776
10	3130	-1.758
11	3080	-1.742
12	3030	-1.727

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TABLE I. Ballistic equation terms (Cont.)

Term	Definition
R	Radar Range Voltage: 4mV/meter
$\dot{R}$	Radar Range Rate Voltage: -32 mV/meter/sec
θ_E	Elevation Lead Angle = $\frac{\theta_A}{25.43}$
L^3_E	Elevation Lead Angle Cubed = $(\theta_A/25.43)^3$
θ_D	Air Density:

Switch Position	AD (Gain factor)
.7	-.1281
.75	-.1148
.8	-01016
.85	-.0886
.9	-.0757
.95	-.0630
1.00	-00511
1.05	-.0394
1.1	-.0280
1.15	-.0178
1.2	-.0082

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3.5.1.1.2.2 Magnet current - target outbound - radar mode.

$$V_{IM} = A_M \left[\left[\frac{-.0236L^2 - 13.188}{R} + \frac{14.314}{R^2} - \frac{8.8315}{R^3} \right) + R \left(\frac{.5551}{R} - \frac{1.337}{R^2} \right)^{-1} \right] - 3.938A_D + .15$$

3.5.1.1.3 Torque current generation. The torque current generator shall be capable of providing a controlled dc torque motor current of variable magnitude to the sight torque motor coil, the current range being between approximately zero and equal to or more than 500 milliamperes (mA).

3.5.1.1.4 Torque current equation. The generator shall supply the appropriate torque motor current required to implement the specified sight superelevation and sight correction terms as defined in the following equation for solution of V_{IT} $'_{IT}$ is the voltage present at the input of the torque current supply module.

3.5.1.1.4.1 Torque current.

$$'_{IT} = (+)*2.840x10^{-2}* (L_E*L^2*V_{IM}^{2903}) (+)* 4.861x10^{-2}$$

$$*(L^3_E*V_{IM}^{2.03}) -6.952x10^{-3}*R - 2.006X10^{-2*R} + 1.859xl0^{-2} \cos 0_E$$

* (+) sign is (+) for a positive lead angle elevation and (-) for a negative lead elevation

3.5.1.1.5 Measurement of magnet and torque currents. Magnet and torque currents, which have pulse type wave shape, shall not be measured as currents; rather, a voltage measurement across the current

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3.5.1.1.5.1 Magnet current. The relationship between ^{v}IM from Table II and the feedback voltage of the magnet load is:

$$^{v}IM \text{ Feedback Voltage} = -\left[(.1346\ ^{v}IM - .0202) \pm 0.3\%\right] \pm .0017 \text{ volt}$$

3.5.1.1.5.2 Torque current. The relationship between V_{IT} from Table II and the feedback voltage of the torque motor load is:

$$^{'}IT \text{ Feedback Voltage} = -\left[\frac{(1.1102\ V_{IT})}{13} \pm 1\%\right] \pm .0038 \text{ volt.}$$

3.5.1.1.6 Accuracy of V_{TM} and V_{IT}.

3.5.1.1.6.1 Magnet current voltage. The ^{v}IM tolerance shall be ± 2 percent of VIM for all test problems of Table II.

3.5.1.1.6.2 Torque current voltage. The V_{IT} tolerance shall be 1.5 percent of absolute conformity based on 13 volts maximum for all problems of Table II.

3.5.1.2 Secondary performance characteristics.

3.5.1.2.1 Input power/internal signals. The generator shall be designed to operate as specified herein when the following input power is supplied and internal signal voltages are present.

3.5.1.2.1.1 Input power.

a. +24 Volts dc - +24 volts dc nominal (22 volts minimum to 30 volts maximum) at 1.5 amps maximum to be received from the control panel.

b. +18.35 Volts dc - +18.3 ± 1.0 volts dc at 1.2 amperes maximum to be received from the inverter.

c. 280 Volts ac - 280 volts peak-to-peak ± 5.0 percent, 400 Hz ± 1.0 percent at 1.5 amperes maximum to be received from the inverter. The 280-volt source shall be center-tapped, and the ac output shall be a quasi-square wave that contains 8.33 percent off-time between each half-cycle.

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TABLE II. Theoretical values of magnet and torque module inputs

Problem Number	L^2T (L=lambda)	L^2E JL=lambda)	CosE	Generator Inputs Density Switch Position	Muzzle Switch Position	RANGE	Range rate	Theoretical module inputs ^vIM	$'IT$
1	0° Ov	0° OV	0° -15V	100	1	500m 2V	-155m/s - 5V	7.9323	-.2853
2	0° Ov	0° Ov	0° -15V	1.0	6	500m 2V	-155m/s -5V	7.6139	-.2853
3	0° Ov	0° Ov	0° -15V	1.0	12	500m 2V	-155m/s - 5V	7.2211	-.2853
4	0° Ov	0° Ov	0° -15V	1.0	6	500m 2V	-275m/s -8.9V	7.4388	-.2575
5	0° Ov	0° Ov	0° -15V	1.0	6	1500m 6V	-155m/s - 5V	3.8555	-.3656
6	0° Ov	0° Ov	0° -15V	1.0	6	1500m 6V	-275m/s 8.9V	3.9722	-.3377
	10° -2.4V	+10° -2.4V	0° -15V	1.0	6	1500m 6V	+155m/s + 5V	3.1518	+.0445

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TABLE II. Theoretical values of magnet and torque module Inputs (Continued)

Problem Number	Generator inputs L²T (L=lambda)	L²E (L=lambda)	CosE	Density Switch Position	Muzzle Switch Position	Theoretical module inputs Range	Range rate	ʸI M	‘IT
8	0° OV	0° Ov	0° -15V	.9	4	1600m 6.4V	-200m/s -6.48V	3.8776	-.3630
9	15° -5.4V	-5° -.6V	0° -15V	1.0	6	2000m 8V	-155m/s - 5V	3.0846	-.6700
10	0° OV	0° OV	0° -15V	.7	6	2500m 10V	-155m/s - 5V	2.9304	-.4459
11	0° OV	0° Ov	0° -15V	1.2	6	2500m 10v	-155m/s 5V	2.4597	-.4460
12	0° Ov	Ov Ov	0° -15V	1.0	6	2500m 10v	-275m/s -8.9V	2.7179	-.4180
13	0° Ov	0° OV	30° -12.99V	1.0	6	2500m 10V	-155m/s - 5V	2.6269	-.4080
14	0° OV	-20° -9.6V	0° -15V	1.0	6	2500m 10v	275m/s 8.9V	2.6822	-1.8934
Test Points	J7-D	J7-E	J7-A			J7-X	J7-AA	J7-G	J7-H

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3.5.1.2.1.2 Internal signal voltages.

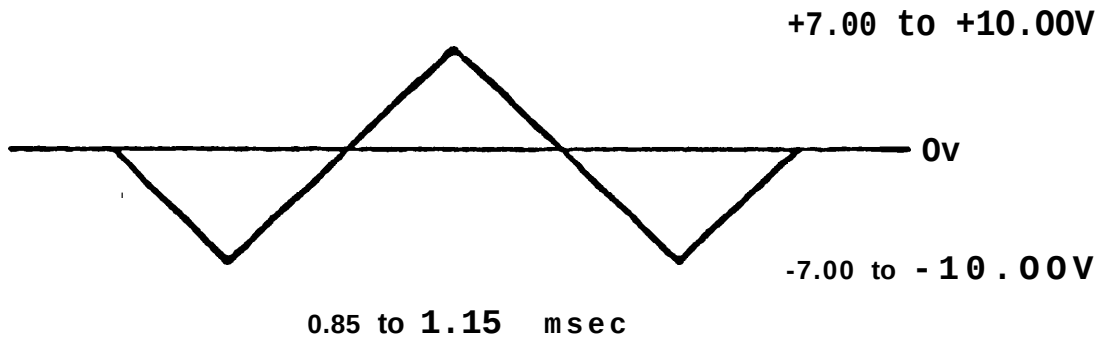
a. +15 Volts dc. +15 ± 0.075 volts dc at 1.33 amperes maximum derived from a de-de power supply within the generator.

b. -15 Volts dc. -15 ± 0.075 volts dc at 1.33 amperes maximum derived from a de-de power supply within the generator.

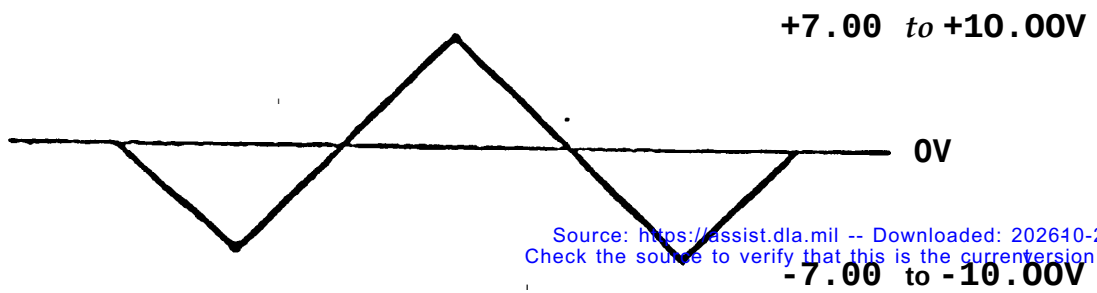
-7 Volts dc. -7 ± 0.001 volts de is a reference voltage measured on the Mode Scaling Board (P/N 12000011).

d. +7 volts dc. +7 ± 0.02 volts dc is a reference voltage measured on the Range Series Board (P/N 12000016).

e. Phase 1. This voltage is generated on the Mode Scaling Board (P/N 12000011) and shall have the following characteristics:



f. Phase 2. This voltage is generated on the Computation Control Board (P/N 12000012), and shall have the following characteristics:



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3.5.1.2.1.2.1 Time delay (10 see). Provision for a six to 14 second time delay shall be required after the application of primary input power to the generator. This circuit will inhibit the ± 15 volt dc supply from being applied to the generator circuitry and reticle lamp excitation power, until after the time out is completed. This delay is required during the following modes of operation: Radar, Manual, Test and External Mode.

TABLE III. V_{IM} Dependency on air density and muzzle velocity settings

Vary air density			Vary muzzle velocity		
Air density switch	Muzzle velocity switch	V_{IM} +2%	Air density switch	Muzzle velocity switch	V_{IM} +2%
.7	6	4.1908	.95	1	4.0895
.75	6	4.1385	.95	2	4.0585
.8	6	4.0865	.95	3	4.0275
.85	6	4.0353	.95	4	3.9965
.9	6	3.9844	.95	5	3.9654
.95	6	3.9345	.95	6	3*9345
1.00	6	3.8076	.95	7	3.9034
1.05	6	3.8415	.95	8	3.8705
1.10	6	3.7966	.95	9	3.8395
1.15	6	3.7565	.95	10	3.8046
1.20	6	3.7187	.95	11	3.7736
			.95	12	3.7746

CONTROL SETTINGS

TARGET INBOUND in RADAR MODE
Lead Angle Traverse (L_T) = O degrees = ± 0.05 Vdc (Test Point 20)
Lead Angle Elevation (L_E) = O degrees = ± 0.05 Vdc (Test Point 19)
Cosine E = O degrees = -15 Vdc (Test Point 16)
RANGE = 6V (Test Point 64)
RANGE RATE = -6V (Test Point 67)
Lock-on Switch to ON
VIM (Test Point 49)

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TABLE IV. Test connector (J7)

J7 Pin	Associated connector Pin	Function
A	JIC, J3U	115 volts, 400 Hz input
B	JIB, J3D	115 volts, 400 Hz return
c	Jlc, Jig, J2N	+24 Volts dc (10-sec time delay (K23) excitation
D	J3L	O to -15 Volts dc from L_T^2 potentiometer
E	J3V	O to -15 Volts dc from L_E^2 potentiometer
F	J6B	Radar lock-on
G	J6W	Electronic countermeasures signal
H	J6K	-10 Volts dc (inbound range rate reference)
J	J6J	+10 Volts dc (outbound range rate reference)
K	J6e	+20 Volts dc (range reference)
L	J5A, J6h	Radiate command (from foot switch)
M	J1X, J6Y	Radar operate
N	J2F	Azimuth o A
P	J2H	Azimuth O B
R	J2G	Azimuth O C
S	J2R	Elevation o A
T	J2S	Elevation o B
U	J2J	Elevation o C
V	J2K	IT Feedback
W	J2D	I_M Feedback
X		Ground
Y		-15 volts dc
Z		+15 volts dc
a	J3B	Cos E potentiometer
b		Ready-to-fire lamp
c	J3X	$'E^3 I_m^2$
d	J1s	Sight cage input (K24 excitation)
e	JIA	Ready-to-fire (K27 excitation)
f		Range greater than 2200 meters
g		^{v}IM
h		'IT

NOTE : L=lambda for function column of J7 Pin D,E,c,n,EE and FF

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TABLE IV. Test connector (J7) (cent)

J7 Pin	Associated connector Pin	Function
i		Air density reference (-3.9375)
j		Muzzle velocity (VIM 5th Term)
k		Muzzle velocity (VIM 5th Term)
m		$1/R^n$ (VIM 1st Term)
n		$.LE^2/R$ (v_{IM} 4th Term)
p		Superelevation signal
q		Divider reference
r		R
s		R/R^2
t		0^1
u		-R/R
v		Calibration, low
w		Calibration, high
x		Range
y		Range, buffered (0.8333 Range)
z	JIZ	Acquisition time delay (K14) excitation
AA		Range rate
BB	J3Y	$+K1_M2$
cc		Acquisition time delay
DD		f (I_M) buffered
EE		$-K1_M2$ L 2
FF		L^2 Signal
GG		Serrisoid waveform voltage
HH		Sight cage signal

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3.5.1.2.2 Functional provisions.

3.5.1.2.2.1 Manual controls. Provisions for manual controls shall include the following:

a. Muzzle velocity. This control shall permit introduction of data representing muzzle velocities between 3030 and 3580 feet per second in 50-feet-per-second increments. See Table III for VIM dependency on muzzle velocity settings.

b. Air density. This control shall permit introduction of data representing air density ratios between 0.7 and 1.2 in increments of 0.05. See Table III for VIM dependency on air density settings.

3.5.1.2.2.2 Acquisition time delay. Provision shall be made for an acquisition time delay circuit which shall be effective in the radar mode of operation.

a. This circuit shall derive a time interval to terminate the sight correction component of the torque current. The time interval shall commence upon radar lock-on and terminate as a function of radar range so that t(seconds) = 0.186 V_R - 0.345 $\pm$ 0.25 seconds as measured at connector pin J7-CC (Table IV).

b. This circuit shall derive a time interval to provide a relay closure to increase the turret rate. The time interval shall commence upon radar lock-on and terminate as function of radar range so that t(seconds) = 0.186 V_R -0.345 $\pm$ 0.25 seconds as measured at connector pin J7-Z (Table IV).

c. This circuit shall derive a time interval to terminate magnet current. The time interval shall commence upon radar lock-on and terminate as a function of range so that t(seconds) = 0.186 V_R -0.345 $\pm$ 0.25 seconds as measured at connector pin J7-W (Table IV). The time as measured from magnet current shall have a tolerance as follows:

Range	Tolerance
R less than or equal to 1500	$\pm$ 0.25 sec
R greater than 1500, less than or equal to 2000	$\pm$ 0.50 sec
R greater than 2000, less than or equal to 2500	$\pm$ 0.75 sec
R greater than 2500, less than or equal to 3000	$\pm$ 1.0 sec

where R = range in meters.

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3.5.1.2.2.3 Modes of operation. The generator shall perform as specified in the following paragraphs.

3.5.1.2.2.3.1 Radar mode. The generator, when in the radar mode, shall use radar range and range rate information for generation of magnet and torque current. The electrical cage position of magnet current output shall be under control of the lock-on signal furnished by the radar. Table II delineates representative test points for this mode.

3.5.1.2.2.3.2 Test mode. When the generator is in the test mode, the magnet and torque currents shall be generated with fixed values, and the current magnitudes shall be measured. If the measured values are within the design tolerance, a test complete ("good when lit") indicator signal shall be sent to the control panel. The range fixed voltage shall be 6.4 volts, and the range rate input voltage shall be -6.841 volts. Muzzle velocity and air density switch settings shall be set as follows:

Muzzle velocity	Air density
4	.95

3.5.1.2.2.3.3 Manual mode. In the manual mode, the generator shall use manually-set target range and range rate information from the control panel for generation of magnet and torque current values. The electrical cage position of magnet current shall be under control of sight cage signal input from the control panel. The ready-to-fire indicator signal to the sight shall be energized at all times in this mode.

3.5.1.2.2.3.3.1 Magnet current - manual mode.

$$V_{IM} = A_M \left[\left[\frac{-.0295^2 - 18.58}{R} + \frac{29.09}{R^2} - \frac{17.24}{R^3} \right) + R \left(\frac{.2798}{R} - \frac{.8971}{R^2} \right) \right) - 3.937A_D + .15$$

3.5.1.2.2.3.3.2 Torque current - manual mode.

$$'_{IT} = (+)^{*}2.840 \times 10^{-2} (L_E^2 VIM^{2003}) \quad (+)^{*}4.861 \times 10^{-2} (L_E^3 V_{IM})^{2*03}$$

$$-9.262 \times 10^{-3} R + 6.95 \times 10^{-2} - 1.605 \times 10^{-2} R + 1.86 \times 10^{-2} \text{CosO E}$$

a (+) sign is (+) for a positive LEAD ANGLE ELEVATION and (-) sign for a negative LEAD ANGLE ELEVATION

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3.5.1.2.2.3.4 External mode. The generator shall provide the following conditions in external mode:

a. The generator shall not provide magnet current for output to the sight.

b. The generator shall provide a fixed value of torque current equal to 19.529 milliamperes at 39 ± 6 millivolts.

c. The electrical cage position of the magnet current shall be under control of the sight case signal input from the control panel.

d. The ready-to-fire indicator signal to the sight shall be under manual (External Range Control) control through electrical interface with the control panel.

3.5.1.2.2.3.5 Ground mode. In the ground mode, the generator shall not provide magnet and torque currents.

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TABLE V. Manual mode: open looP sight current generator response

RANGE (VOLTS) (M)	(TP-64)	TARGET SPEED (KNOTS)	(VOLTS) (TP-67)	LEAD ANGLE (VOLTS) (TP-19-20)	AIR DENSITY SETTING	MUZZLE VELOCITY SETTING	(volts) (TP-49)	(volts) (TP-50)
500	2.5	300	3.75	0.0	1	6	7.616 ± 0.300	−0.286 ± 0.065
500	2.5	600	0	0.0	1	6	70403 ± 00300	-0.252 ± 0.065
1000	5.0	300	3*75	000	1	6	5.112 + 0.250	-0.326 + 0.065
1000	5.0	600	0	0.0	1	6	5.251 ± 0.250	-0,292 ± 0.065
1500	7.5	300	3.75	0.0	1	6	3,851 + 0.200	-0.367 + 0.065
1500	7.5	600	0	0.0	1	6	3.997 ± 0.200	-0.332 ± 0.065
2000	10.0	300	3.75	0.0	1	6	30109 + 0.200	−00407 + 0.065
2000	10.0	600	0	0.0	1	6	3.239 ± 0.200	-0.372 ± 0.065
2500	12.5	300	3*75	0.0	1	6	2.623 ± 0.200	-00447 + 0.065
2500	12.5	600	0	0.0	1	6	2.738 ± 0.200	-0.412 ± 0.065

1/ Gun elevation angle = 0° (CoSE = -15 Vdc)

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3.5.1.2.3 Input signals.

3.5.1.2.3.1 Radar range. A dc signal voltage with an amplitude proportional to radar target range shall be supplied to the generator. Scaling of range input shall be linear so that 20 volts represents a target range of 5000 meters ± 0.1 percent. Input range voltage shall not exceed 20 volts. Minimum radar range shall be representative of a 200 meter target. In the absence of any detectable target, radar range voltage shall assume a fixed voltage representative of a 1600 meter target.

3.5.1.2.3.2 Manual range. In the absence of radar range data, the generator shall be capable of accommodating manually-set target range data from the control panel. This signal shall be scaled for 3000 meters per unit gain, a dc voltage of +15 volts representing 3000 meters and a dc voltage of zero representing zero meters.

3.5.1.2.3.3 External range. when neither radar range nor manual range data is available to the generator, it shall be capable of accommodating manually-set (external) target range data as previously described under 3.5.1.2.2.3.4.

3.5.1.2.3.4 Radar range rate. A dc signal voltage with an amplitude proportional to radar target range rate shall be supplied to the generator. Scaling of range rate input shall be linear so that negative 10 volts represents a target rate of 308.66 meters per second inbound and positive 10 volts represents a target rate of 308.66 meters per second outbound, ± 3.0 meters per second. Input rate voltage shall not exceed 10 volts absolute value. In the absence of any detectable target, radar range rate shall assume a fixed voltage representative of a target rate of -200 meters per second (inbound).

3.5.1.2.3.5 Manual range rate. In the absence of radar range rate data the generator shall be capable of accommodating manually-set target speed from the control panel. This signal shall be scaled for 600 knots per unit gain, a dc voltage of ●7.5' volts representing zero knots, and a dc voltage of zero volts representing -600 knots.

3.5.1.2.3.6 Muzzle velocity. Circuit provisions shall accommodate the output signal of the muzzle velocity control specified in 3.5.1.2.2.1.a.

3.5.1.2.3.7 Air density. Circuit provisions shall accommodate the output signal of the air density control specified in 3.5.1.2.2.1.b.

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3.5.1.2.3.8 Elevation shaft angle. A dc signal (cosine E) representing gun mount elevation shaft angle shall be received from the elevation potentiometer. Minus 15 Volts dc shall correspond to 0 degrees gun elevation, and near zero volts dc shall correspond to +80 degrees gun elevation.

3.5.1.2.3.9 Elevation lead angle. Signals L_E, L_E2, L_E3, representing angle determined by the shaft angle of radar elevation trunions, shall be received from the range-only radar. For $(L_E)^2$, -15 Volts dc shall correspond to 25 degrees lead angle and zero volts dc shall correspond to 0 degrees lead angle.

3.5.1.2.3.10 Traverse lead angle. Signal L_T2, representing lead angle determined by the shaft angle of radar traverse trunions, shall be received from the range-only radar. For L_T2, -15 volts dc shall correspond to 25 degrees lead angle and zero volts dc shall correspond to 0 degrees lead angle.

3.5.1.2.3.11 Sight cage. A sight cage signal (under manual option) from the control panel shall activate the electrical cage position of sight magnet current (approximately 700 miniamperes).

3.5.1.2.3.12 Radar radiate. A radar radiate command signal shall be received from the radar radiate foot switch.

3.5.1.2.4 Output signals.

3.5.1.2.4.1 Magnet Current. A dc magnet current value shall be supplied to the lead-computing sight with the value and limits shown in Table 1I, and for any given dynamic range and range rate signals below the rate slope input of 2.0 V per second. Magnet current load shall be as shown in Figure 1. The sight electrical cage position of magnet current shall be approximately 700 milliamperes.

3.5.1.2.4.2 Torque current. A dc torque current value shall be supplied to the lead-computing sight with the value and limits shown in Table II when the radar range and range rate signals are available below the rate slope input of 2.0 V per second. Torque current load shall be as shown in Figure 2.

3.5.1.2.4.3 sight ready-to-fire indicator. The ready-to-fire indicator signal to the sight shall be energized under the following conditions. The indicator signal requirement shall be 24 volts nominal at 0.18 amperes nominal.

Manual mode. In manual mode, the ready-to-fire indicator signal Shall be energized at all times.

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b. External mode. In the external mode, the ready-to-fire indicator signal shall be controlled by an externally initiated signal from the control panel.

c. Radar mode. In the radar mode, activation of the ready-to-fire indicator signal shall depend on the following conditions:

(1) Radar lock-on signal present.

(2) Radar range signal shall correspond to less than 2200 meters for targets inbound or less than 1600 meters for targets outbound.

(3) Lead angle (L^2) signal shall represent less than $(25)^2$ degrees.

(4) Elevation angle (cosine E) signal shall represent less than 80 degrees.

(5) Acquisition time delay shall be complete.

d. Radar mode, electronic countermeasures (ECM) activity. When a radar-jammed signal is received from the radar, then the sight ready-to-fire signal shall be interrupted at a 2.5 Hz $\pm$ sO percent rate. (The ECM condition shall override any other ready-to-fire signal.)

3.5.1.2.4.4 Test complete. A test complete indicator signal shall be sent to the control panel upon successful completion of the test specified in 3.5.1.2.2.3.2. The indicator signal requirement shall be 24 volts nominal at 0.18 amperes nominal.

3.5.1.2.4.5 Radar radiate. A nominal +24 volt radar radiate command signal shall be sent to the range only radar when a corresponding signal is received from the radar radiate foot switch.

3.5.1.2.5 Discrete input/output signals.

3.5.1.2.5.1 Radar signals. Provision shall be made for generation of the following discrete signals. (These signals are to be independently activated by closure of associated relay contacts in the range only radar.)

a. Radar operational (Radar Ready).

b. Radar-jammed (electronic countermeasures (ECM) activity).

Radar lock-on. When the radar lock-on signal is absent, the electrical cage position of sight magnet current (3.5.1.2.2.3.1 and 3.5.1.2.4.1) shall be effective.

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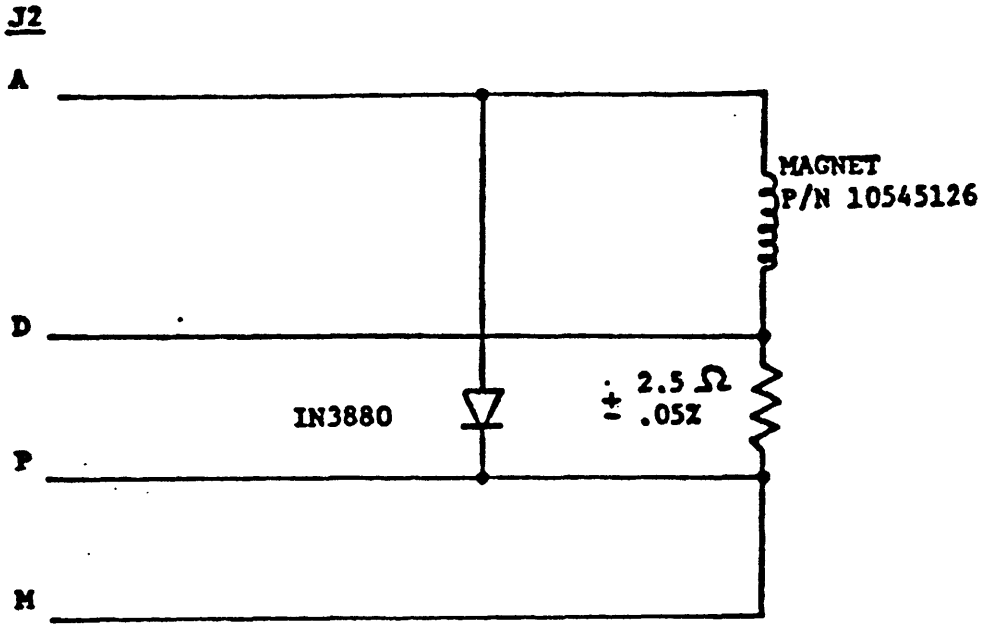


FIGURE 1. Magnet current load.

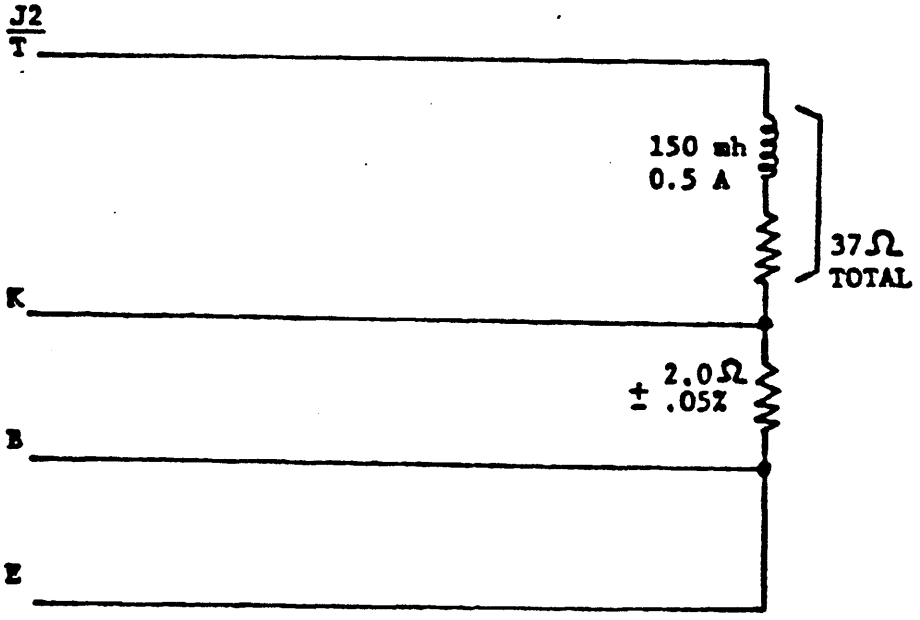


FIGURE 2. Torque current load.

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3.5.1.2.5.2 Control panel signals. provision shall be made for generation of the following discrete signals. (These signals are to be independently activated-by associated relay or switch-closures in the control panel.)

a. Action switch command
b. Ready-to-fire command
c. Antenna servo command
d. Sight cage command
e. Manual mode command
f. External mode command

3.6 Component definition.

3.6.1 Interface requirements.

3.6.1.1 Schematic arrangement.

3.6.1.1.1 Mechanical. The mechanical interface of the generator with other components in the weapon system shall be as follows:

a. M163A1 per 12007500
b. M167A1 per 12007501

3.6.1.1.2 Electrical. The electrical interface of the generator with other components in the weapon system shall be in accordance with Figure 3 and Tables VI through XI.

3.6.1.2 Detailed interface definition.

3.6.1.2.1 Input power. Input power interface shall be in accordance with the requirements of 3.5.1.2.1.1.

3.6.1.2.2 Input signals. Input signal interface shall be in accordance with the requirements of 3.5.1.2.3.1 through 3.5.1.2.3.5 and 3.5.1.2.3.8 through 3.5.1.2.3.12.

3.6.1.2.3 Output signal. Output signal interface shall be in accordance with the requirements of 3.5.1.2.4.

3.6.1.2.4 Discrete input/output signals. Discrete input/output signal interface shall be in accordance with the retirements of 3.5.1.2.5.

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3.7 Design and construction.

3.7.1 General design features.

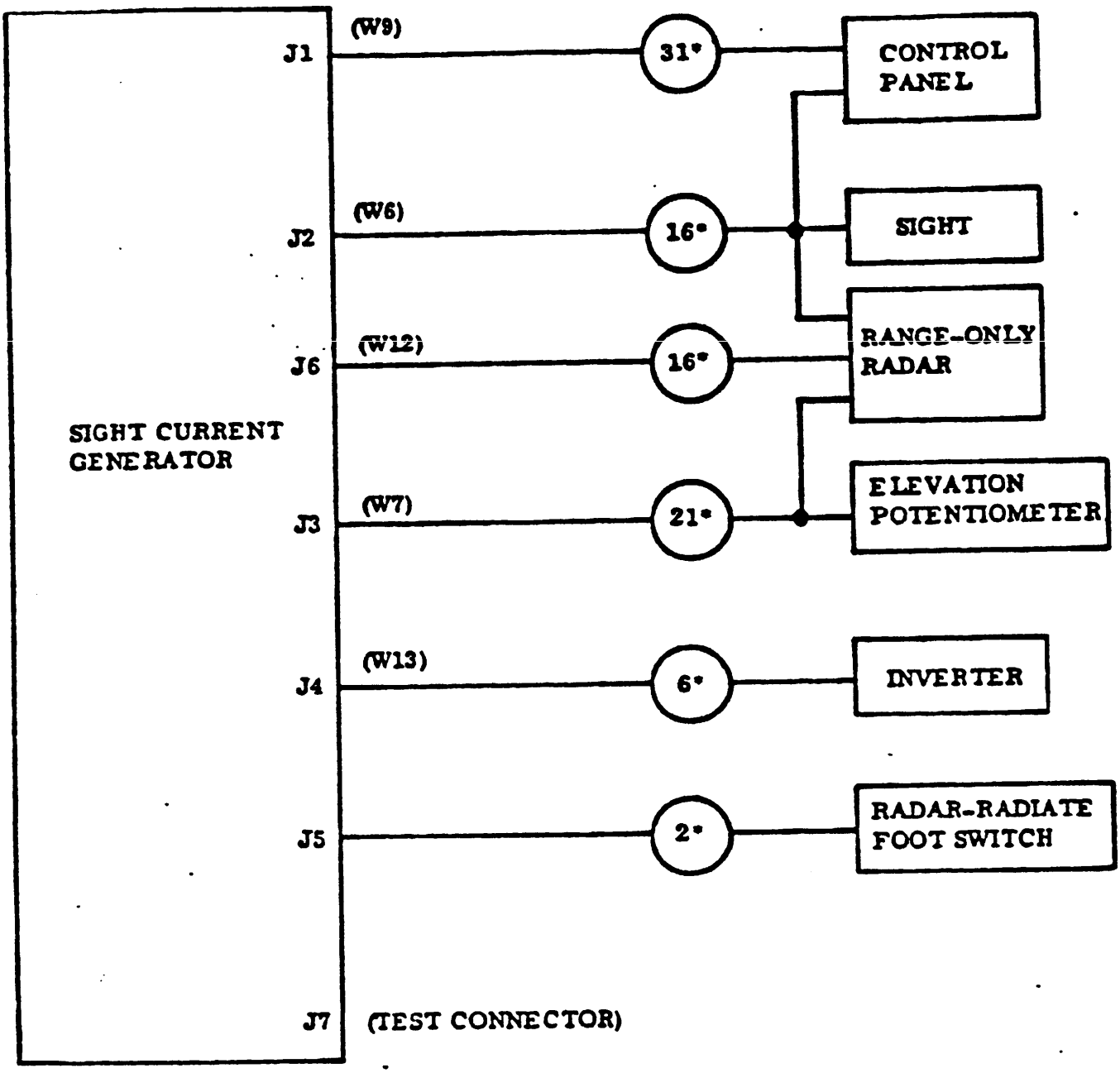
3.7.1.1 Dimensions. All dimensions shall be in accordance with 12000239.

3.7.1.2 Weight. The weight of the generator shall not exceed 36 pounds.

3.7.1.3 Electrical connectors. Electrical connectors shall be mounted on one side of the generator opposite to the controls specified in 3.5.1.2.2.1.

3.7.1.4 Vibration isolators. The generator mounting brackets shall be designed to accommodate four vibration isolators (Part Number 10545465) between the brackets and the mounting surface.)

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NOTES:
1. *Designates wires per cable.

FIGURE 3. Schematic diagram of electrical interface.

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TABLE VI. Sight current generator connector pin assignment, control panel connector (J1)

PIN	FUNCTION
A	Ready-to-fire (K27) excitation
B	115 volts, 400 Hz return
C	115 volts, 400 Hz input
L,N	Shield ground (J1B, C; J3D, U)
K	e_E > 80° NC switch input
M	-15 volts dc
P,R,U, n	Ground
S	Manual range input
T	Manual range rate input
V	Test lamp return
W, d	Test lamp excitation
X	Radar operate
Z	Acquisition time delay (K14) excitation
f	+15 Volts dc
g	External range ready-to-fire
h	f (I_M) Return
j	External mode (K25) excitation
k	Action switch command (K22) excitation
s	Sight cage input (K24) excitation
	The following connections shall be required for the stated conditions:
p	Ground mode (+24-volt dc input)
b,e,r	Radar mode (+24-volt dc input)
E,e,r	Manual mode (+24-volt dc input)
m,r	External mode (+24-volt dc input)
b,c,e,r	Test mode (+24-volt dc input)
	Action switch control
A	+24-Volt dc input
e,k	(Connected through Action Switch on Control Panel)

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TABLE VII. Lead-computing sight connector (J2)

PIN	FUNCTION
A	I^* output
P	I_m Return
B	I_t Return
K	$*_t$ Feedback
D	I_m Feedback
M	I_m Return
E	I_t Return
T	$'_t$ output
F	Azimuth OA
H	Azimuth OB
G	Azimuth OC
R	Elevation OA
S	Elevation OB
J	Elevation OC
L	Ready-to-fire lamp return
N	Ready-to-fire lamp excitation

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TABLE VIII. Range-only radar connector (J3)

PIN	FUNCTION
A	-15 Volts dc to Cos- Potentiometer
B	Cos_E Potentiometer (Wiper)
G	$-KI^2$ L^2 to L_E Potentiometer Cup 1
H	$+KI_m^2$ L L_E to L_E Potentiometer Cup 1 (Wiper)
J	$+KI_m^2$ L^2 to L_E Potentiometer Cup 1
E,F	-15 Volts dc to L_E^2 Potentiometer Cup 2
v	0 to +15 Volts dc to L_E^2 Potentiometer Cup 2 (Wiper)
w	$-Ki_m$ to L_E Potentiometer Cup 3
x	$'_E{}^3$ I_m to L_E^3 Potentiometer Cup 3 (Wiper)

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TABLE VIII. Range-only radar connector (J3)

PIN	FUNCTION
Y	$+K I_m^2$ to L_E^3 Potentiometer Cup 3
L	0 to +15 Volts dc to L_T^2 Potentiometer (Wiper)
M,z	-15 Volts dc to L_T^2 Potentiometer
D	115 Volts 400 Hz return
u	115 Volts 400 Hz input
b	Shield ground for pins D, U
K,R,a,c	Ground

NOTE: L = lambda for function column of Pins G, H, J, E, F, V, W, x, Y, L, M, z

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TABLE IX. Inverter connector (J4)

PIN	FUNCTION
A	+18.35 Volts dc input
B	18.35 Volts dc return
E	Shield ground for pins F, G, H
F	280-Volt quasi square wave center tap
G,H	280-Volt p-p, 400 Hz, quasi square wave input

TABLE X. Radar radiate foot switch connector (J5)

PIN	FUNCTION
A	Radiate command
B	+24 Volts dc through action switch (K22)

TABLE XI. Range-only radar electronics connector (J6)

PIN	FUNCTION
J	+10 Volts dc (outbound range rate reference)
K	-10 Volts dc (inbound range rate reference)
N	+10 to -10 Volts dc (range rate signal)
e	+20 Volts dc (range reference)
L	O to +20 Volts dc (range signal)
v,b,d,r	Ground
B	Radar lock-on (N.C. Contact)
C	+15 Volts dc
W	Electronic countermeasures signal
X,g	+24 Volts dc (through action switch K22)
h	Radiate command
Y	Radar operate

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3.8 **First article.** When specified in the contract or purchase order, a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3).

3.9 **Soldering.** All soldering shall be in accordance with the general requirements of MIL-STD-2000 in addition to detail requirement (task) F of paragraph 1.2 of MIL-STD-2000.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of Sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 General provisions. The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Special sampling inspection (see 4.5).

4.2.1 Inspection conditions. Unless otherwise specified, all examinations and tests shall be conducted at the following standard ambient conditions:

	Temperature	+50 deg to +90 deg F.
b.	Relative humidity	90% RH Max.
c.	Pressure	20 to 32 inches of Hg.

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4.3 First article inspection.

4*3.1 First article submission. The contractor shall submit a first article sample as designated by the contracting officer consisting of one generator for inspection in accordance with 4.3.2.

4.3.2 Inspections to be performed. First article assemblies, components and test specimens may be subjected by the Government to any or all of the examinations and tests specified in Table XII and Table XIII herein and to any or all requirements of the applicable drawings.

4.3.3 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of any assembly, component or test specimen to comply with any of the stated requirements.

4*4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term "inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which they are drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.

4.4.2 Examinations and tests.

4.4.2.1 Components, assemblies and subassemblies. All components, assemblies and subassemblies shall be inspected in accordance with the inspection provisions contained in this specification and the Quality Assurance Provisions (QAP) listed in the Technical Data Package (TDP). In the absence of QAPs, the applicable quality assurance provisions of MIL-F-13926 shall apply.

4.4.2.2 Final acceptance inspection. Subsequent to first article approval, examination and tests related to Section 3 herein shall be performed in accordance with the sampling plans specified in 4.4.2.3 herein. Examination and test for packaging and marking shall be in accordance with section 5 herein. The tabulated classification of defects in 4.4.2.3 shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

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4.4.2.3 Functional tests. The tests in Table XII shall be performed on a 100% basis.

TABLE XII. Functional tests
(100% Inspection)

CHARACTERISTIC		REQUIREMENT	TEST PROCEDURE
Critical:		None	N/A
Major:			
101.	Electrical	3.4	4.7.2
102.	Performance	3.5	4.7.3
Minor:		None	N/A

4.5 Special sampling. One generator out of the first 50 produced and one out of each 100 produced-thereafter shall be selected at random. The sample Shall have met the requirements and tests of Table XII and shall then meet the tests in Table XIII. Unless otherwise specified in the contract, the items shall be returned to the lot upon successful completion of testing.

TABLE XIII. Special sampling

CHARACTERISTIC		REQUIREMENT	TEST PROCEDURE
Critical:		None	N/A
Major:			
103	Vibration	3.3.1	4.7.1.1
104.	Shock	3.3.2	4.7.1.2
105.	Altitude	3.3.3	4.7.1.3
106.	Temperature	3.3.4	4.7.1.4
107.	Humidity	3*3.5	4.701.5
108.	Rain	3.3.6	4.7.1.6
Minor:		None	N/A

4.5.1 Failure of sample. Should any one item of a special sampling fail to meet any of the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections to all production since the last test as well as to current production has been accomplished in accordance with the provisions of the contract.

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4.6 Inspection equipment.

4.6.1 Government-furnished inspection equipment. where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

4.6.2 Contractor-furnished inspection equipment.

4.6.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment identified by drawing number in specifications or QAPs forming a part of the contract shall be supplied by the contractor in accordance with the design specified. The contractor may, however, propose alternatives to Government designs for approval in accordance with 6.3.

4.6.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the "Methods of Inspection" specified in 4.7 of this specification and with the component inspection procedures specified in 'Examination" and 'Test Facilities" requirements of MIL-F-13926 whenever Government designs are not specified (see 4.6.2.1). Since tolerance of test equipment is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment. Contractor inspection equipment designs shall be submitted for approval in accordance with 6.3.

4.6.2.3 Test equipment. In conjunction with 4.6.2.1 and 4.6.2.2, the test equipment listed in Table XIV shall be used in the performance of the applicable test methods and procedures as specified in 4.7.

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TABLE XIV. Test equipment list

ITEM NUMBER		DESCRIPTION
1.	Power supply	o - 30 Vdc at 20A
2*	Inverter	10545536
3.	Oscilloscope	Tektronix T922
4.	Voltmeter, digital	Fluke 8600A
5.	Test fixture	SCG/TX-13; see Table XV for test fixture parts list, see Figure 4 for test fixture schematic.
6.	Power supply	± 10 Vdc
7.	Power supply	± 20 Vdc

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TABLE XV. Test fixture parts list

DESIGNATION	NOMENCLATURE AND DESCRIPTION
A1C1	Capacitor, ●047 uf, 50 VDC
A1CR1, 2, 3 and 8	Diode, 1N4003
AlCR5	Diode, 1N3064
A1CR7	Diode, 1N968B
A1Q1	Transistor, 2N1595
A1Q2 thru Q4	Transistor, 2N2904
A1R8	Resistor, 2K ohms, l/4W ± 1%
A1R9	Resistor, 7.5K ohms, l/8W ± 1%
A1R10	Resistor, 25K ohms, l/8W ± 1%
A1R12	Resistor, 2.0 ohms, lW ± 0.05%
A1R13	Resistor, 51 ohms, l/2W ± 1%
A1R14	Resistor, 383 ohms, lW ± 1%
A1R15	Resistor, 2K ohms, l/2W + 5%
A1R16	Resistor, 470 ohms, l/2W ± 5%
A1R18 and 20	Resistor, 180 OhmS, lW ± 5%
A1R19 and 21	Resistor, 180 ohms, l/4W ± 5%
A1R22	Resistor, 100 ohms, l/4W ± 5%
A2	Magnet assembly, Drawing 10545126
A3	Torque motor assembly, Drawing 10545170 or Figure 2
DS1 thru DS10 and 12	Lamp, miniature, GE 327
DS11	Lamp, miniature, GE 330
F1	Fuse, type 3AG, 5A
K1 and K2	Relay 6PDT, 5A Cont. 26 VDC Coil
P1	Connector, plug, MS3106F28-21S
P2	Connector, plug, MS3106F20-29P
P3	Connector, plug, MS3126F16-26P
P4	Connector, plug, MS3126F16-8S
P5	Connector, plug, MS3106F12S-3P
P6	Connector, plug, MS3126F20-41S
P7	Connector, plug, MS3126F22-55P
P1 (Inverter)	Connector, plug, MS3116F14-12S
P2 (Inverter)	Connector, plug, MS3116F16-8S
R1 and R3	Potentiometer, IRC-7300-lK
R2 and R4	Potentiometer, Clarostat 73JA-10K
R5	Potentiometer, Drawing 10545437
R6 and R7	Potentiometer, Helipot 7603-R25K-(1)
R11	Resistor, 2.5 ohms, 5W + 0.05%
S2, 12 and 13	Switch, toggle SPST, 5 amp min.
S14, 15, 17, 18 and 19	Switch, toggle SPDT, 5 amp min.
S9	Switch, toggle DPST, 5 amp min.
S16	Switch, toggle TPDT, 5 amp min.

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TABLE xv. Test fixture parts list (cont.)

DESIGNATION	NOMENCLATURE AND DESCRIPTION
S4, 6, 7 and 10	Switch, momentary SPST, 5 amp min.
S5	Switch, momentary DPDT, 5 amp. min.
S8	Switch, momentary SPDT, 5 amp min.
Sll	Switch, rotary 4 pole, 5 position (break before make)
TP1 thru TP92	Tip jack, H.H. Smith -1501 or equal
XDS1 thru XDS1O and 12	Lampholder, Dialco 162-8430-0931-502
XDSll	Lampholder, Dialco 162-8430-0932-502
XF1	Fuseholder, Littelfuse 342014

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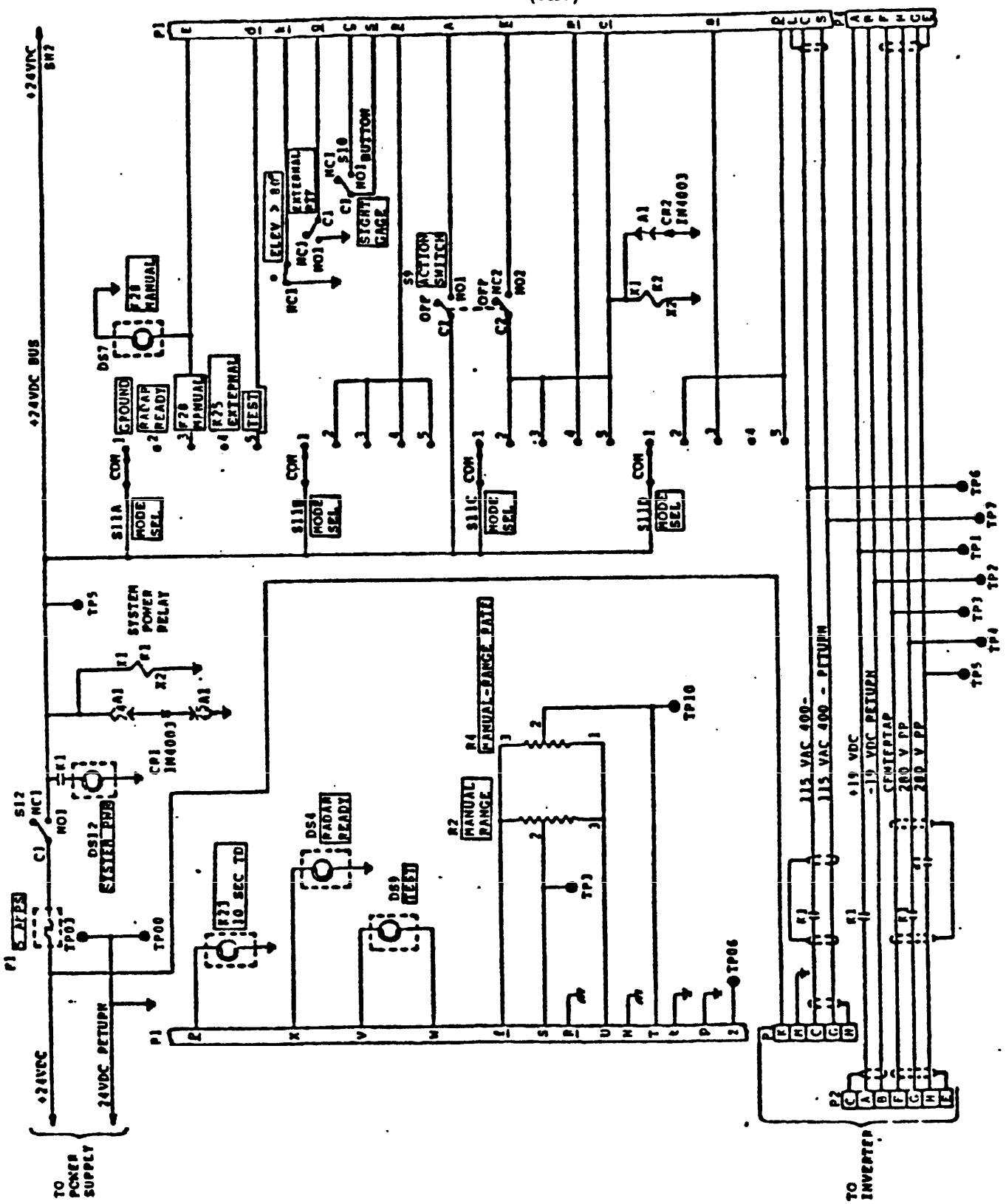


FIGURE 4. Test fixture schematic (1 of 3)

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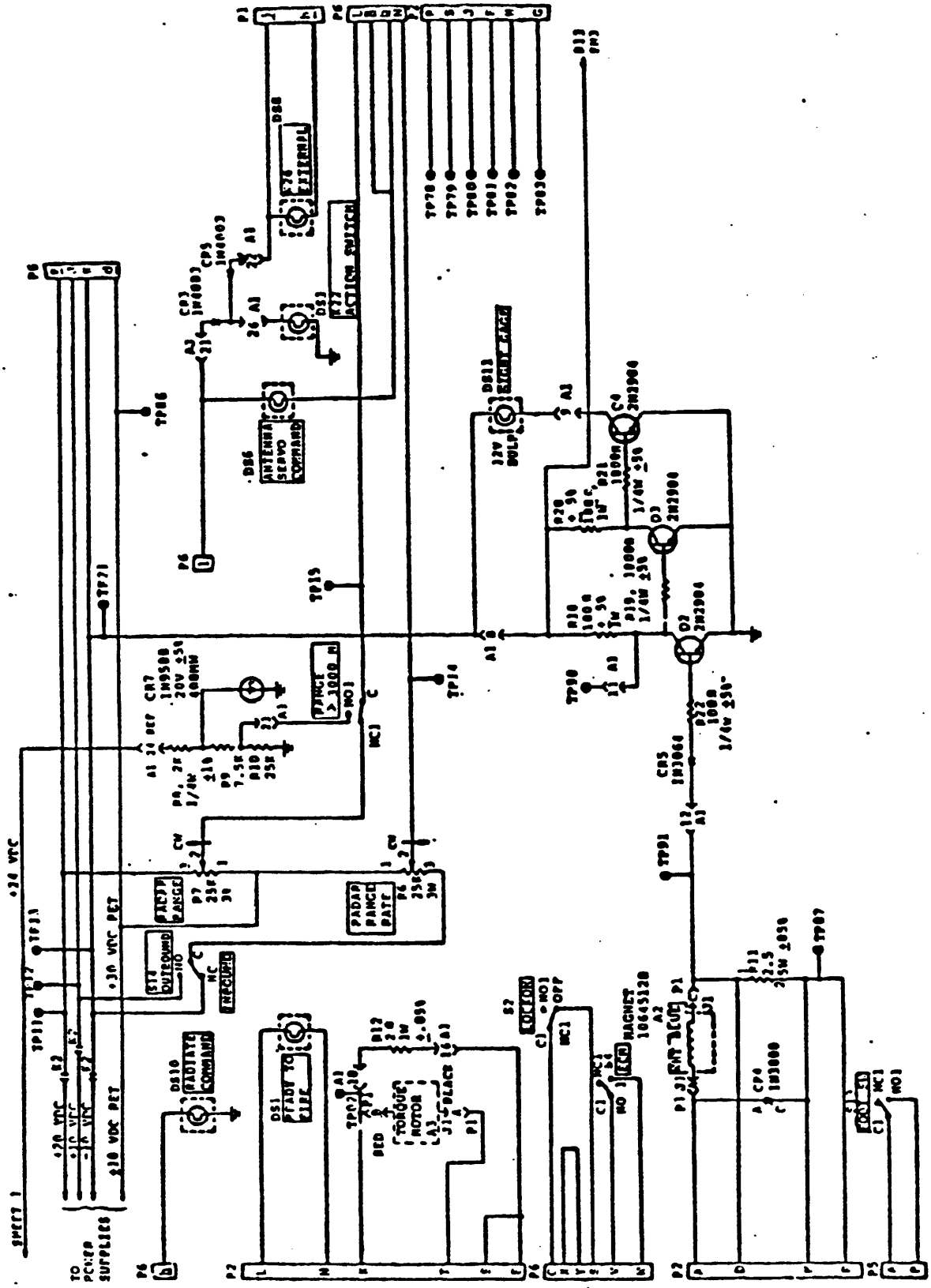


FIGURE 4. Test fixture schematic (2 of 3)

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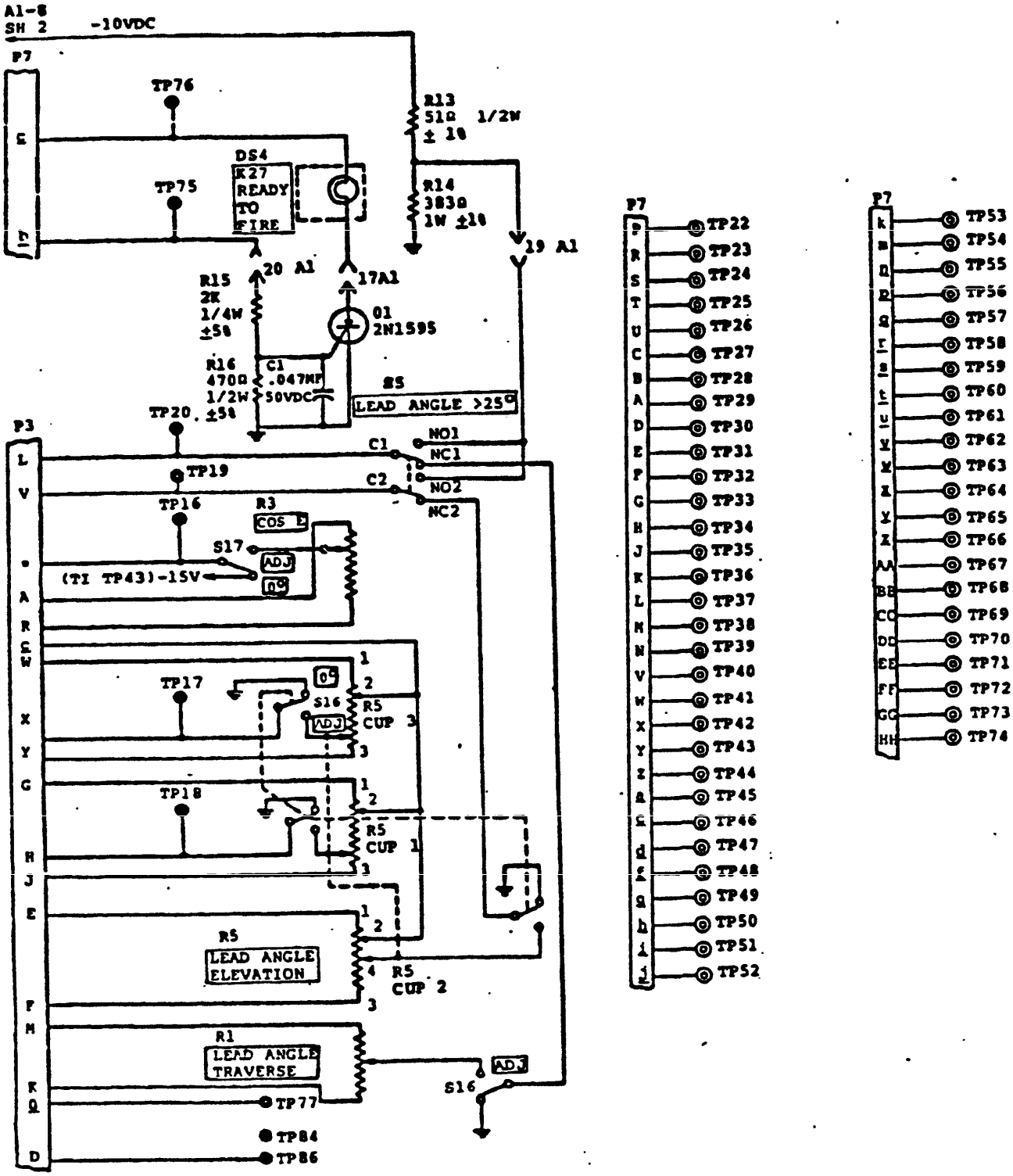


FIGURE 4. Test fixture schematic (3 of 3).

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4.7 Methods of inspection.

4.7.1 Environmental. Environmental tests shall be conducted in accordance with the test conditions, general test performance guidance, and individual test methods of MIL-STD-810 except as specified herein. Test facilities, chambers and apparatus used in conducting the tests contained in this specification shall be capable of meeting the conditions required. During environmental testing, the internal signal voltage requirement, 4.7.3.6.3 thru 4.7.3.6.6, shall be omitted from the performance requirements of 4.7.3.

4.7.1.1 Vibration. The test shall be performed as specified in Method 514, Category 8, procedure I, o MIL-STD-810fas modified by 3.3.1 herein. In Step 1 of Procedure I only steps 4 through 8 of paragraph 11-2.1 apply. The generator shall be attached to the vibration generator directly or with a fixture, and securely held by its normal means of attachment. After vibration, the generator shall be tested as specified in 4.7.2 and 4.7.3 herein to determine compliance with 3.3.1.

4.7.1.2 Shock. This test shall be performed as specified in Procedure I, Section II of Method 516 of MIL-STD-810 except that:

a. The post test functional check (Step 5 of Procedure I) need not be conducted until completion of all shock impulses.

b. The shock pulses shall be 15 g's ± 10% amplitude and 11 ms ± 10% duration.

c. The maximum g shall occur at 5.5 milliseconds ± 10%.

d. The shock measuring system shall have a flat frequency response within ± 10% from 20 to 2,000 hertz. After the application of all the shock impulses, the generator shall be tested as specified in 4.7.2 and 4.7.3 to determine compliance with 3.3.2.

4.7.1.3 Altitude (operating and non-operating). This test shall be performed as specified in Procedures II and III, Method 500 of MIL-STD-810 . The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.3.

4.7.1.4 Temperature.

4.7.1.4.1 High (operating and non-operating) This test shall be performed as specified in Procedures I and II, Method 501 of MIL-STD-810 with constant temperature extremes as specified in 3*3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

4.7.1.4.2 Low (operating and non-operating) This test shall be performed as specified in procedures I and II, Method 502 of MIL–STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

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4.7.1.5 Humidity (non-operating) This test shall be performed in accordance with procedure 111, Method 507 of MIL-STD-810 except that in Step 7, checkout shall be performed at the conclusion of-the test only, with the item at standard ambient conditions.

4.7.1.6 Rain (nonoperating) This test shall be performed as specified in Procedure I of MIL-F-13926.

4.7.2 Electrical. Continuity shall be inspected by standard measuring equipment (SME) and visual examination for compliance with 3.4.

4.7.3 Performance. The selected generator shall be tested as follows: Connect power supply (Table XIV, item 1), inverter (Table XIV, item 2), oscilloscope (Table XIV, item 3), and voltmeter (Table XIV, item 4) to the test fixture (Table XIV, item 5) in *accordance* with Figure S. Selection of mode and connection of inputs, outputs and power shall be in accordance with Table VI.

4.7.3.1 Preset controls. All control settings are on the test fixture with the exception of the AIR DENSITY and MUZZLE VELOCITY controls which are on-the generator under test. All measurements are made on the test fixture unless otherwise stated.

Preset controls are as follows:

RADAR RANGE - 8.0 ± 0.5 Vdc as measured at TPIS

RADAR RANGE RATE – control to mid-point

INBOUND OUTBOUND - switch to INBOUND

LEAD ANGLE TRAVERSE - switch TO “O”

LEAD ANGLE ELEVATION - switch to "0"

COSINE - switch to ‘O”

LOCK-ON - switch to OFF

ACTION SWITCH - switch to OFF

FOOT SWITCH - switch to OFF

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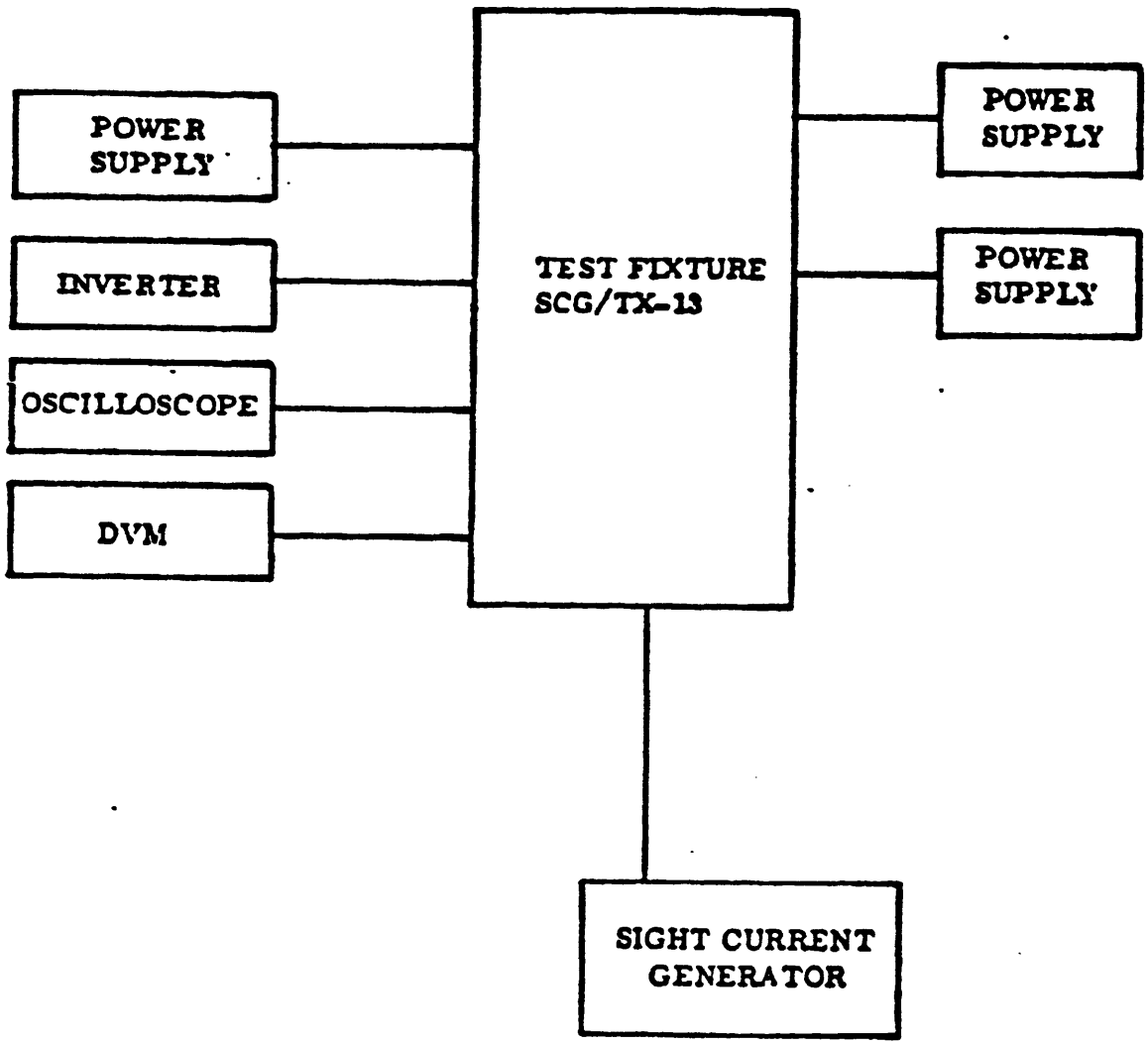


FIGURE 5. Test set up.

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4.7.3.2 System power. Set the MODE switch to GROUND and operate the SYSTEM POWER ON/OFF switch to ON. observe that the SYSTEM POWER indicator lights.

4.7.3.3 Radar ready mode indicators. Set MODE switch to RADAR READY: observe that the RADAR READY indicator lights; observe that after-a six to 14 second delay K23 10 SECOND TIME DELAY indicator lights and the SIGHT CAGE indicator also lights. The above indications are in accordance with 3.5.l.2.5.la, 3.5.1.2.1.2.1, and 3.5.1.2.2.3.1.

4.7.3.4 Action switch. Set ACTION SWITCH to ON and observe that the following Indicators light:

K22 ACTION SWITCH
ANTENNA SERVO COMMAND

K27 READY-TO-FIRE

The above indications are in accordance with 3.5.l.2.5.2a, b and c.

4.7.3.5 Foot switch. Set the FOOT SWITCH to ON and observe that the RADIATE COMMAND indicator lights. This indication is in accordance with 3.5.1.2.3.12 and 3.5.1.2.4.5.

4.7.3.6 Internal signal voltages.

4.7.3.6.1 +15 vdc. Measure the dc voltage between test points 44(+) and 42(-), in accordance with 3.5.l.2.l.2a.

4.7.3.6.2 -15 Vdc. Measure the dc voltage between test points 43(+) and 42(-), in accordance with 3.5.l.2.l.2b.

4.7.3.6.3 -7 vdc. Measure the dc voltage between test points A2TP4 (mode scaling) and 42(-), in accordance with 3.5.1.2.1.2c; adjust A2R6 as required.

4.7.3.6.4 +7 Vdc. Measure the dc voltage between test points A16A2TP4 (range series) and 42(-), in accordance with 3.5.l.2.l.2d.

4.7.3.6.5 Phase 1. Using an oscilloscope, measure the voltage and waveform at test points A2TP3 (mode scaling) and 42(-), in accordance with 3.5.l.2.l.2e.

4.7.3.6.6 Phase 2. Using an oscilloscope, measure the voltage and waveform at test points A3TP5 (computation control) and 42(-), in accordance with 3.5.l.2.l.2f.

4.7.3.7 Lock-on. Set the LOCK-ON switch to ON and observe that the SIGHT CAGE indicator goes out, in accordance with 3.5.1.2.5.lc. observe that the READY-TO-FIRE indicator lights after a one to two second time delay, in accordance with 3.5.1.2.4.3c(5).

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4.7.3.8 Range greater than 3000M. Depress RANGE GREATER THAN 3000M switch and observe that the READY-TO-FIRE indicator goes out, in accordance with 3.5.1.2.4.3c(2).

4.7.3.9 Elevation greater than 80°. Depress ELEVATION GREATER THAN 80° switch and observe that the READY-TO-FIRE indicator goes out, in accordance with 3.5.1.2.4.3c(4).

4.7.3.10 Lead angle greater than 25°. Depress LEAD ANGLE GREATER THAN 25° switch and observe that the READY-TO-FIRE indicator goes out, in accordance with 3.5.1.2.4.3c(3).

4.7.3.11 ECM. Depress ECM switch and observe that the READY-TO-FIRE indicator blinks, in accordance with 3.5.l.2.4.3d and 3.5.l.2.5b.

4.7.3.11.1 ECM. Depress simultaneously the ECM switch and the ELEVATION GREATER THAN 80° switch, and observe that the READY-TO-FIRE indicator blinks, in accordance with 3.5.l.2.4.3d.

4.7.3.12 Sight cage. Depress SIGHT CAGE switch and observe that the SIGHT CAGE indicator lights, in accordance with 3.5.l.2.5.2d and 3.5.1.2.3.11.

4.7.3.13 Ready to fire. Measure the dc voltage at test points 15(+) and 42(-) and adjust the RADAR RANGE control until the READY-TO-FIRE indicator goes out. This voltage shall read +8.4 to +9.2 Vdc, to take in the various tolerance stackup which occur, in accordance with 3.5.1.2.4.3c(2). Continue to adjust RADAR RANGE for a voltage of between +8.999 and +9.001 Vdc. Set LOCK-ON switch to OFF and remove digital voltmeter lead. observe SIGHT CAGE indicator lights, in accordance with 3.5.1.2.4.1.

4.7.3.14 Acquisition time delay functions.

4.7.3.14.1 Torque current termination. Using an oscilloscope, measure the pulse duration between test points 69(+) and 42(-) when LOCK ON switch is set to ON, the duration shall be in accordance with 3.5.l.2.2.2a. Adjust RADAR RANGE voltage at test points 64(+) and 42(-) between +8.999 and +9.001 Vdc. (This voltage represents a range of 2250 meters).

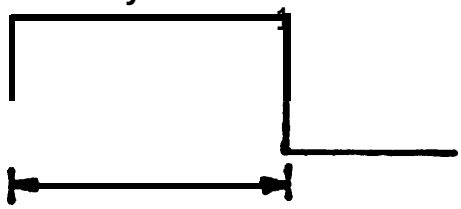
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NOTE

To repeat pulse display, set LOCK ON switch to OFF and then to ON as often as necessary. Pulse amplitude voltages are approximately as shown below.

+10 to +15 VDC

-0.4 VDC



1.1 to 1.6 sec

a. Repeat the above procedure for each of the following RADAR RANGE voltage settings:

(1) 4 volts (1000 meters) ATD = 0.15 to 0.65 sec
(2) 7 volts (1750 meters) ATD = 0.71 to 1.2 sec
(3) 11 volts (2750 meters) ATD = 1.4 to 2.0 sec

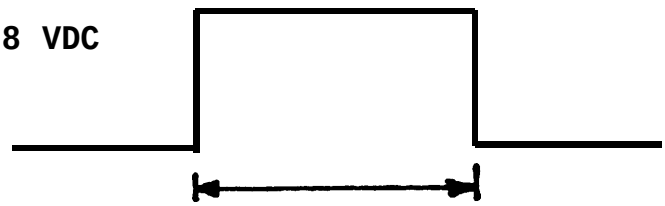
4.7.3.14.2 Relay closure. Using an oscilloscope measure the pulse duration between test points 66(+) and 42(-) when the LOCK ON switch is set to ON, the duration shall be in accordance with 3.5.l.2.2m2b. Adjust the RADAR RANGE voltage at test points 64(+) and 42(-) between +8.999 and +9.001 Vdc.

NOTE

To repeat pulse display, set LOCK ON switch to OFF and then to ON as often as necessary. Pulse amplitude voltages are approximately as shown below.

+26 to +28 VDC

o VDC



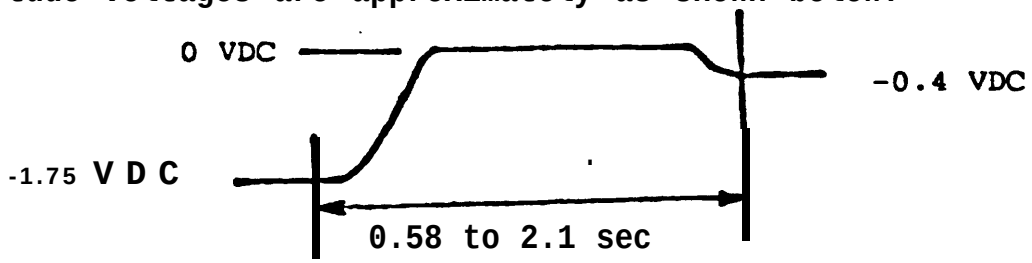
1.1 to 1.6 sec

4.7.3.14.3 Magnet current termination. Using an oscilloscope measure the pulse duration between test points 41(+) and 42(-) when the LOCK ON switch is set to ON, the duration shall be in accordance with 3.5.1.2.2.2c. Adjust RADAR RANGE voltage at test points 64(+) and 42(-) between +8.999 and +9.001 Vdc. (This voltage represents a range of 2250 meters).

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NOTE

To repeat pulse display, set LOCK ON switch to OFF and then to ON as often as necessary. Pulse amplitude voltages are approximately as shown below.



4.7.3.15 Manual mode. Rotate MODE switch to K26 MANUAL and observe the following indicator conditions in accordance with 3.5.7.2.5.2e and 3.5.1.2.1.2.1:

a. K26 MANUAL indicator lights.
b. K23 10 SEC TD and RADIATE COMMAND indicators go out.
c. K23 10 SEC TD and READY-TO-FIRE indicators light after a 6 to 14-second delay.

4.7.3.15.1 Settings and measurements. Set the MUZZLE VELOCITY switch to position 12 and the AIR DENSITY switch to position 1.0 on the generator under test.

a. Set MANUAL RANGE SWITCH to ADJUST. Adjust the MANUAL RANGE control to a voltage level of +2.50 + 0.02 Vdc as measured between test points 64(+) and 42(-) in accordance with 3.5.1.2.3.2 (+2.5 Vdc represents a manual range of 500 meters).

b. Set MANUAL RANGE RATE SWITCH to ADJUST. Adjust the MANUAL RATE control to a voltage level of +3.75 ± 0.02 Vdc as measured between test points 67(+) and 42(-) in accordance with 3.5.1.2.3.5 (+3.75 Vdc represents a manual range rate of 300 knots).

Measure the dc voltage between test points 49(+) and 42(-) (V_{IM} input) in accordance with 3.5.1.2.2.3.3.1. The voltage level shall be +7.260 ± 0.300 Vdc.

d. Measure the dc voltage between test points 50(+) and 42(-) (v_{IT} input) in accordance with 3.5.1.2.2.3.3.2. The voltage level shall be - 0.284 ± 0.065 vdc.

4.7.3.15.2 Manual mode Table V. Repeat 4.7.3.15.1 using the settings of Table V.

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4.7.3.16 V_{IM} and V_{IT} feedback.

4.7.3.16.1 Settings and measurements. Set the MODE SWITCH to RADAR READY and LOCK ON switch to OFF. Observe the READY-TO-FIRE, K26 MANUAL and K23 10 SEC TD indicators go out. After 6 to 14 seconds the SIGHT CAGE and K23 10 SEC TD indicators shall light.

a. Set the MUZZLE VELOCITY switch to position 4 and the AIR DENSITY tO position .95 on the generator under test. Set the LOCK ON switch to ON and the INBOUND/OUTBOUND switch to INBOUND.

b. Measuring the voltage between test points 64(+) and 42(-), adjust the RADAR RANGE control to obtain a voltage level of +6.40 ± .05 Vdc .

c* Measuring the voltage between test points 67(+) and 42(-), adjust the RADAR RANGE RATE control to obtain a voltage level of -6.48 ± .05 Vdc.

d. Measuring the voltage between test points 20(+) and 42(-), set the LEAD ANGLE TRAVERSE switch to 0°; obtain a voltage level of 0 ± .05 Vdc, in accordance with 3.5.1.2.3.10.

e. Measuring the voltage between test points 19(+) and 42(-), set the LEAD ANGLE ELEVATION switch to 0°; to obtain a voltage level of 0 ± .05 Vdc, in accordance with 3.5.1.2.3.9.

f. Measuring the voltage between test points 16(+) and 42(-), set the COSINE switch to 0° to obtain a voltage level of -15.000 ± .075 vdc, in accordance with 3.5.1.2.3.8. Tolerance shall increase to ± .150 Vdc during temperature testing.

4.7.3.16.2 V_{IM} feedback. Measure and record the voltage between test points 49(+) and 42(-) (VIM). The voltage level shall be +3.827 Vdc ± 2%.

a. Measure and record the voltage between test points 41(+) and 42(-) and compute V_{IM} feedback in accordance with 3.5.1.1.5.1 using the V_{IM} recorded level above. The measured VIM feedback level shall be within the required limits of the computed level.

4.7.3.16.3 V_{IT} feedback. Measure and record the voltage between test points 50(+) and 42(-) (V_{IT}). The voltage level shall be -.362 + .195 vdc.

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a. Measure and record the voltage between test points 40(+) and 42(-) and compute V_{IT} feedback in accordance with 3.5.1.1.5.2 using the V_{IT} recorded level above. The measured V_{IT} feedback level shall be within the required limits of the computed level.

4.7.3.17 Radar mode.

4.7.3.17.1 Settings and measurements. Set the MUZZLE VELOCITY switch to position 6 and the AIR DENSITY switch to 1.0 on the generator under test.

Measure the voltage between test points 64(+) and 42(-) and adjust the RADAR RANGE control for a voltage level of +8.00 ± .05 vdc .

b. Measure the voltage between test points 67(+) and 42(-) and adjust the RADAR RANGE RATE control for a voltage level of -8.90 ± 0.05 Vdc.

c. Measure the voltage between test points 20(+) and 42(-) and set the LEAD ANGLE TRAVERSE switch to 0°, to obtain a voltage level of O ± 0.05 Vdc.

d. Measure the voltage between test points 19(+) and 42(-) by first setting the LEAD ANGLE ELEVATION switch to ADJUST then adjusting the LEAD ANGLE ELEVATION control past -15° to obtain a voltage level of -9.60 ± 0.05 Vdc, in accordance with 3.5.1.2.3.10.

e. Measure the voltage between test points 16(+) and 42(-). Set the COSINE switch to 0°, to obtain a voltage level of -15.000 ± 0.075 Vdc. Tolerance shall increase to ± .150 Vdc during temperature testing.

4.7.3.17.2 V_{IM} measurement. Measure the voltage between test points 49(+) and 42(-). This voltage level shall be ± 3.170 ± 2% Vdc.

4.7.3.17.3 V_{IT} measurement. Measure the voltage between test points 50(+) and 42(-). This voltage level shall be -2.427 ± 0.195 Vdc .

4.7.3.17.4 Radar mode Table II. Repeat 4.6.3.17.1, 2 and 3 using the settings in Table II.

NOTE

In problem 7 the INBOUND/OUTBOUND switch is set to OUTBOUND. All tolerances are in accordance with 3.5.1.1.6 for VIM and V_{IT} measurements.

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4.7.3.18 External mode.

4.7.3.18.1 Control settings and observations. Set the MODE switch to K25 EXTERNAL and observe that the following indictors go out :

K(23 10 SEC TD)	ANTENNA SERVO COMMAND
K22 ACTION SWITCH	RADIATE COMMAND
	RADAR READY

After six to 14 seconds, the following indicators shall light in accordance with 3.5.1.2.5.2:

K23 10 SEC TD	25 EXTERNAL
K22 ACTION SWITCH	

b. Measure the voltage between test points 40(+) and 42(-) (v_{IT} Feedback). This voltage level shall be 0.039 ± 0.006 Vdc, in accordance with 3.5.l.2.2.3.4b.

Depress EXTERNAL READY-TO-FIRE switch and observe that the READY-TO-FIRE indicator lights in accordance with 3.5.l.2.2.3.4d.

d. Depress the SIGHT CAGE switch and observe that the K25 EXTERNAL indicator goes out in accordance with 3.5.1.2.2.3.4c.

4.7.3.19 Test mode.

4.7.3.19.1 Control settings and observations. Set the ACTION SWITCH and the LOCK ON switches to OFF and set the MODE SWITCH to TEST . observe that the following indicators go out:

K23 10 SEC TD	K27 READY-TO-FIRE
K22 ACTION SWITCH	K25 EXTERNAL

The RADAR READY indicator shall light and after six to 14 seconds, the K23 10 SEC TD and the SIGHT CAGE indicators shall light in accordance with 3.5.1.2.5.lc and 3.5.1.2.1.2.1.

Set the ACTION SWITCH to ON and observe that the SIGHT CAGE indicator goes out and the following indicators light in accordance with 3.5.1.2.5.2.

K22 ACTION SWITCH	ANTENNA SERVO COMMAND
K27 READY-TO-FIRE	RADIATE COMMAND

b. Set the switches and controls on the test fixture and generator under test to the values and positions as indicated in 4.7.3.16.la through f. The TEST indicator shall light in accordance with 3.5.1.2.2.3.2 and 3.5.1.2.4.4.

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4.7.3.20 Air density and muzzle velocity. Set controls and make measurements as delineated in Table 111 which includes representative test points and tolerances in accordance with 3.5.1.2.2.

4.7.3.21 Continuity check. Measure the resistance between the following test points: (The resistance shall be less than 2.5 ohms.)

Test point	Test point
77	89
78	24
79	25
80	26
81	39
82	22
83	23
86	66

4.8 soldering. Soldering processes shall be inspected by certified inspectors to the requirements and acceptance criteria of MIL-STD-2000 as specified in 3.9.

5. PACKAGING

5.1 General. Preservation, packing, and marking shall be in accordance with Packaging Data Sheet PS12000239. The level of protection shall be as specified in the procurement document.

6. NOTES

This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.

6.1 Intended use. The generator is part of (VADS) either the M163A1 or M167A1 Weapon System. The generator, when provided with data for air density, muzzle velocity, angular position, and target range and range rate, will compute the required magnet and torque current for output to a lead-computing sight.

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6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.
b. Applicable stock number.
First Article Sample requirements.
d. Packaging requirements, if other than specified *in* Section 5.
e. Serialization requirements, if applicable.
f. Certification of conformance for each lot of shipment of product.
g. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).

6.3 Submission of contractor inspection equipment designs for approval. Submit two copies of designs as required to: Commander, U.S. Army AMCCOM, ATTN: AMSMC-QAF-I, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research, Development and Engineering Center (ARDEC), may also include drawings prepared by, and identified as, U.S. Army Armament, Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC.

6.5 Subject term (key word) listing.

Fire Control System
Vulcan Air Defense System (VADS) Self-propelled M163A1
Vulcan Air Defense System (VADS) Towed M167A1

6.6 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1285-A721)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-G-60662B (AR)	1 May 1991

3. DOCUMENT TITLE

GENERATOR, SIGHT CURRENT 12000239

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
US ARMY ARDEC STANDARDIZATION OFFICE	(1) Commercial (201) 724-6674	(2) AUTOVON 880-6674
ADDRESS (Include Zip Code) ATTN: SMCAR-BAC-S PICATINNY ARSENAL, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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